

Graphical Abstract

Scientific Title of your plan

Your complete name, Raja Hashim Ali, Your Second Supervisor

Highlights

Scientific Title of your plan

Your complete name, Raja Hashim Ali, Your Second Supervisor

- Research highlight 1
- Research highlight 2

Scientific Title of your plan

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Abstract

Two sentences of background and importance of this topic.

Two sentences defining the gap analysis (work not done in field) and your problem statement.

Two sentences about the methodology of your work.

Three sentences of Results and significance of your results.

One sentence summarizing your contribution.

Keywords: component, formatting, style, styling, insert

1 General instructions

2 This section is a a set of generalized instructions for writing a journal
3 paper. Read these instructions carefully, and delete this section and its text
4 after completing the write up.

- 5 1. **SECTIONS:** You must have exactly the mentioned sections with the
6 given (or more) number of paragraphs.
- 7 2. **TEMPLATE:** Do not delete any text from any of the sections (except
8 General Instructions) in this template. Comment out the text and write
9 your text in the next lines after the commented out text.
- 10 3. **PAGE LIMITS:** Observe the report page limits (minimum 30 or more
11 pages without citations/references).
- 12 4. **PARAGRAPHS:** Each paragraph must consist of 7-10 sentences. A
13 sentence consists of full stop ".", not lines.
- 14 5. **SENTENCES:** Each sentence should start on a new line in latex text
15 (one sentence per line).

- 16 6. **FIGURES:** Make your figures in Microsoft Visio, Canva, DrawIO
17 and/or Powerpoint only. Save them as PDFs. Barcharts and graphs
18 should be saved directly as pdfs (resolution of image should at least
19 be 300 ppi) as soon as they are generated. Save the code (ipynb, R,
20 other code sources) for generating the figures OR save the source file
21 (ppt, canva, visio file) of the original program. Figures with png or
22 jpeg extension, except for where screenshots are the only option, are
23 not to be used.
- 24 7. **SOURCE FILES:** SAVE ALL SOURCE FILES OF FIGURES OR
25 THEIR LINKS IN THE "Source" FOLDER. In case of figures, keep
26 your raw data table also stored as excel file in this repository.
- 27 8. **CODES:** Save all your codes in "Codes" folder in the repository. Also
28 create a GitHub repository with all the code, paste the accessible link
29 to the GitHub repository in a text file, and add that text file in the
30 Codes folder.
- 31 9. **DATASETS:** Save all your datasets in the "Datasets" folder. If dataset
32 is too large (e.g., with images), then save the dataset link in a text file
33 and add this file in the datasets folder.
- 34 10. **TABLES:** Generate all tables in Excel first, then convert them into la-
35 tex tables by using <https://www.tablesgenerator.com> website and past-
36 ing the table into the site along with drawing its borders. Copy and
37 paste this table into latex.
- 38 11. **REFERENCES:** Add references where required. Use bibliography.bib
39 file for generating references and '
40 cite{' command to cite the required work in your write up. Total ref-
41 erences should be between 25 and 50. Use latest references with 80%
42 references later than (not including) 2021. References should prefer-
43 ably be from journals and conferences only (WITH PREFERENCE
44 TO IEEE, Springer, ACM, PloS, Elsevier (ScienceDirect) or MDPI
45 journals/conferences). Preferably no websites, thesis, Researchgate or
46 Arxiv links. Take the bibtex of all references only from Google Scholar,
47 not the site themself.
- 48 12. **MAIN TEXT REF:** All figures and tables should be referred to in
49 the main text.
- 50 13. **CAPTIONS:** All captions for figures and tables should be detailed,
51 and not three to four words. Think of figures with captions to be self
52 explanatory on their own, without the need to read the main text in

document. Of course, the detailed discussion is not part of captions,
 and done in relevant section of main text.
 14. The sample breakdown of your report is shown in Figure 1.

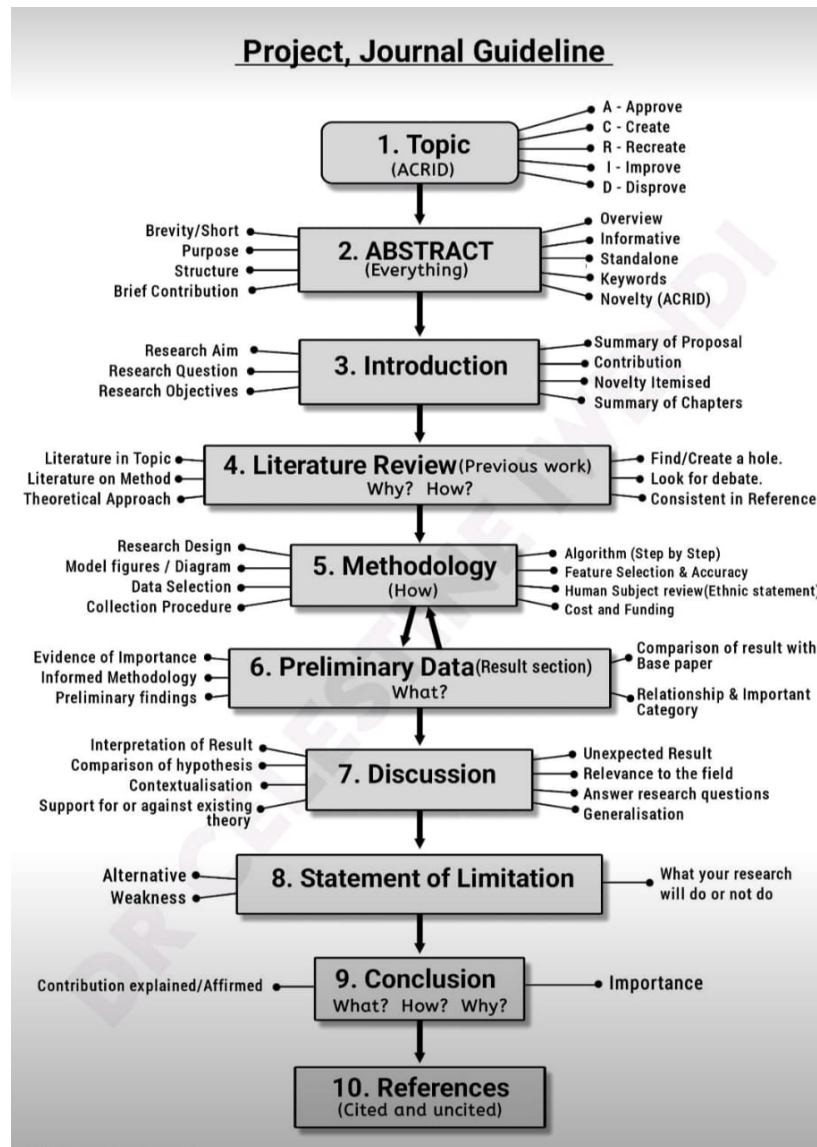


Figure 1: Sample Sections and what to expect in each section.

56 1. Introduction

57 First paragraph on importance and applications of your topic [1]. A figure
58 to either introduce your topic (e.g., for brain tumor analysis, an image of
59 brain indicating a healthy brain versus a brain with tumor highlighted), OR
60 that gives statistical importance/applications of your topic (e.g., for diabetes,
61 an image of world map, showing the spread of diabetes across regions).

62 Second paragraph on introducing the main technique (e.g., deep learn-
63 ing) and its importance in solving your main topic. Also on how this main
64 technique is significant/helpful in solving your topic.

65 1.1. Gap Analysis

66 One paragraph defining what has not been done (Gap analysis).

67 1.2. Research Questions

68 Enumerate as shown below what you are doing in this work (one main
69 and at least two smaller research questions) and your contribution/novelty
70 of your work.

71 1. RQ1: Statement of Research Question 1.

72 2. RQ2: Statement of Research Question 2.

73 3. RQ3: Statement of Research Question 3.

74 1.3. Problem Statement

75 One paragraph that gives the exact problem statement solved in this
76 study.

77 1.4. Novelty of this study

78 One paragraph shedding light on novelty of this study - why your study
79 is unique and what it does to address the gaps identified in the previous
80 paragraphs using one-two liner entry followed by at least three bullet points
81 as shown below.

82 • Novelty 1.

83 • Novelty 2.

84 • Novelty 3.

85 *1.5. Significance of Our Work*

86 One paragraph concluding the section with an overall summary of meth-
87 ods, results, and discussion.

88 **2. Literature Review**

89 One paragraph defining all the techniques that have been used in this
90 field.

91 *2.1. Technique 1 Name*

92 One paragraph for this important technique/method with ideally four
93 references (2 sentences per reference).

94 *2.2. Technique 2 Name*

95 One paragraph for this important technique/method with ideally four
96 references (2 sentences per reference).

97 *2.3. Technique 3 Name*

98 One paragraph for this important technique/method with ideally four
99 references (2 sentences per reference).

100 *2.4. Technique 4 Name*

101 One paragraph for this important technique/method with ideally four
102 references (2 sentences per reference).

103 A table summarizing the characteristics of the existing literature along
104 with the novelty of your proposed work (as shown in Table 1).

105 **3. Methodology**

106 One paragraph with three - four lines summarizing your overall method-
107 ology, with a work flow figure (shown in Fig 2 or in Fig 3).

108 You have the liberty to make your own in Powerpoint, Figma or Canva,
109 but ensure the following things are correctly done in the figure.

- 110 • Use white background for making your workflow figures, and possibly
111 dark colors for arrows, icons, and black/grey text.
- 112 • All steps should be represented by a number in a colored circle with
113 contrasting colors.

Table 1: Literature Review Summary (2024–2025 Studies)

Topic	Reference	Methodology	Dataset	Challenges Addressed	Limitations	Performance Metrics
Attention Mechanisms	Kumain et al. (2024) [2]	Dual-stream attention model for spatio-temporal cues	LRS2	Word-frame separation, robustness	Fusion complexity	F1-score, interpretability [3]
	Chen et al. (2024) [4]	Lightweight TA-TCNN with skip connections	LRW-1000	Reduced size, temporal refinement	Less generalization in noisy input	Accuracy, compactness
	Ghosh et al. (2024) [5]	ViT block appended to TCNN pipeline	AVA-ActiveSpeaker	Improved visual cue extraction	Pretrained dependency on ViT	Recall, attention weights
Sub-word / Phoneme Units	Oghbai et al. (2025) [6]	Viseme-phoneme mapping + TCNN	TCD-TIMIT, OuluVS2	Enhanced phoneme recognition	Complex alignment processing	Phoneme accuracy, WER
	Chen et al. (2024) [4]	Grapheme-aware encoder with CTC loss	CUAVE, LRW-1000	Linguistic ambiguity resolution	Data imbalance issues	WER, robustness
	Xie et al. (2023) [7]	Hierarchical phoneme-viseme embeddings + TCNN	OuluVS2	Ambiguity reduction in mapping	High training time	Phoneme-F1, efficiency
Visual Speech Detection (VSD)	Sakia et al. (2025) [8]	Multi-scale 3D-CNN + TCNN	AVLetters, LRS2-VSD	Speech activity detection accuracy	High compute load	Precision, recall, FAR
	Jiang et al. (2025) [9]	Transformer-TCNN hybrid for endpoint detection	LRW-500	Visual boundary localization	Requires fine annotation	Detection latency, accuracy
	Stern et al. (2022) [10]	Compressed VSD model for low-resource setting	LRW-100	Real-time performance, resource savings	Slight drop in detection precision	F1 score, efficiency tradeoff
Integration of TCNNs into LipNet	Kosarkar et al. (2025) [11]	SyncVSR: 3D-CNN + TCNN with audio-visual alignment	LRS2, LRS3	Improved spatio-temporal modeling	Increased computational cost	WER, data efficiency
	Sakia et al. (2025) [8]	SwinLip: Swin Transformer + TCNN	LRW, LRW-1000	Global lip motion encoding	Requires Swin frontend	Accuracy, speed
	Oghbai et al. (2025) [6]	VALLR: Video Transformer + LLM decoder	LRS2, LRS3	Phoneme-level alignment with LLM	Two-stage inference flow	WER, model complexity
	Ali et al. (2025) [12] Proposed study	Viseme-guided synthesis + TCNN Compressed VSD model for low-resource setting	LRW LRW-100	Robust to varied visual conditions Real-time performance, resource savings	Synthetic data realism gap Slight drop in detection precision	Accuracy in-the-wild F1 score, efficiency tradeoff

- If a main step has substeps, then all substeps are present in a rectangle, with same number as main step but having their own indices (e.g., 1.1 and 1.2 and so on for main step 1, 2.1 and 2.2 and so on for main step 3).
- All steps and substeps must use arrows that point to the next step and so on. The arrows should use the same color at the same level (main step) but a different color for a different level (substeps).
- Use colorful icons to represent each step and subset where possible. Where not possible, use two to three words to name your step/method/technique.
- Use two to three words with each icon. If someone does not understand the icon, then this text will help the reader. But do not use longer texts whatsoever.

- Do not waste space in the figure. Avoid having white spaces as much as possible.
- Ensure that the appropriate (preferably large) font size is selected so that the image is readable when printed.
- Try and maintain symmetry in the figure where needed.
- Sample workflow figures you want to use as sample, which satisfy these conditions are given in the folder “Figures” → “Sample Figures for Reference” → “Workflow” folder. Review them carefully to exactly know what kind of Figure is required and clear here. You will find that all the sample figures follow these major guidelines.

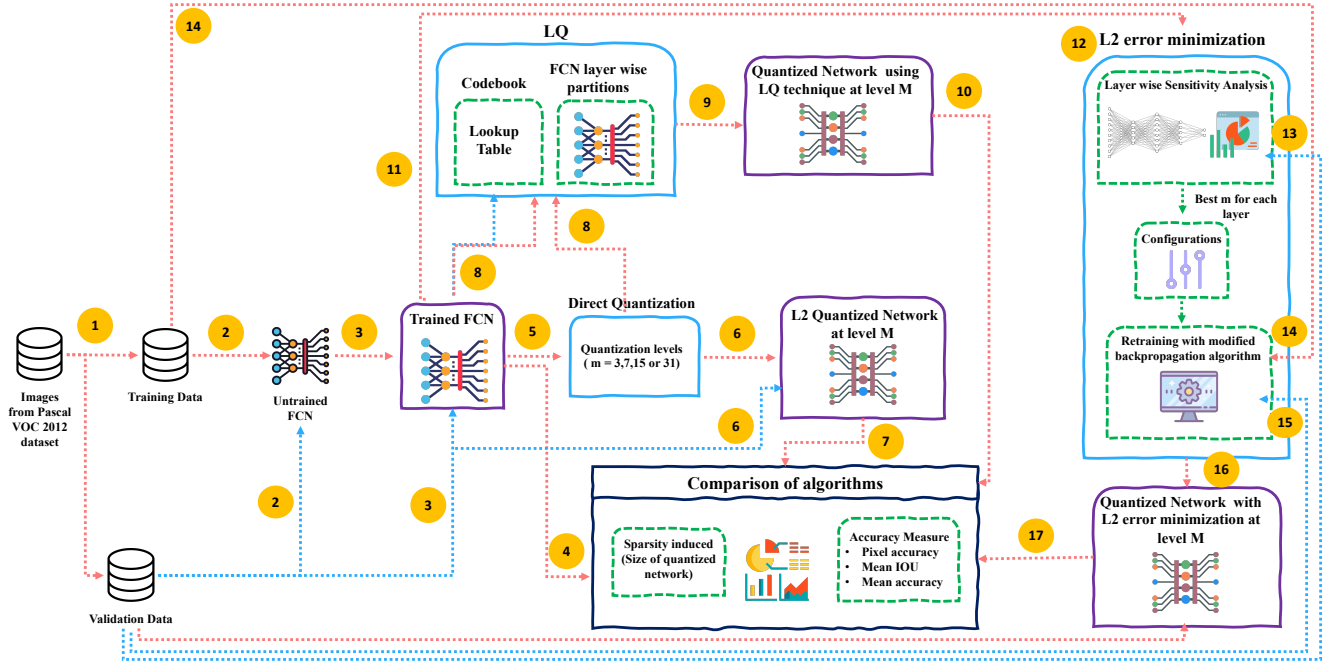


Figure 2: Sample workflow image. Make it in Powerpoint with svg images and save as pdf. Sanity check: Zoom in and pixels should not break.

3.1. Dataset

One paragraph and a figure (or a table) (shown in Fig 4 or in Fig 5) explaining the dataset used in the study. In case of more than one datasets, one paragraph and one figure (or table) for each dataset.

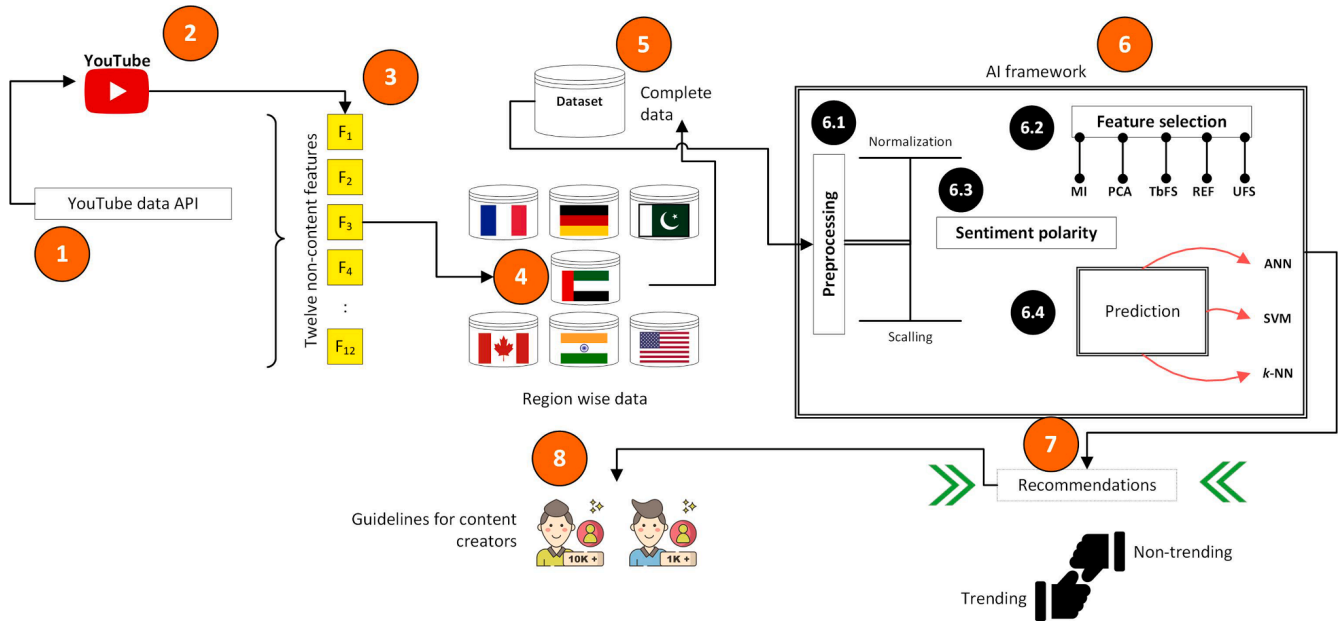


Figure 3: Sample workflow image. Make it in Powerpoint with svg images and save as pdf. Sanity check: Zoom in and pixels should not break.

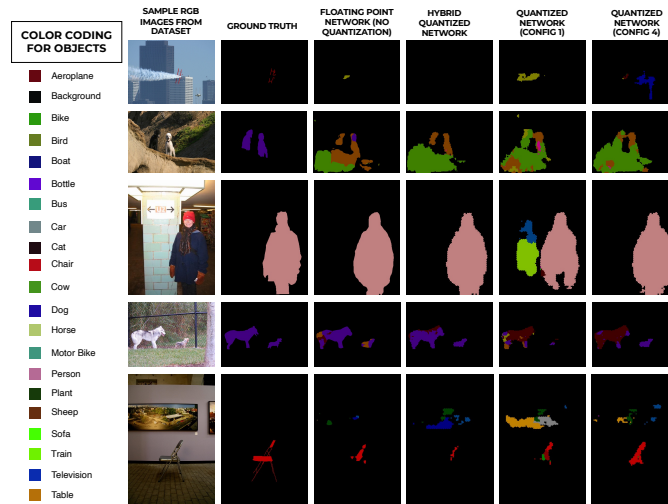


Figure 4: Sample dataset image for images. Make it in Powerpoint with svg images and save as pdf. Sanity check: Zoom in and pixels should not break.

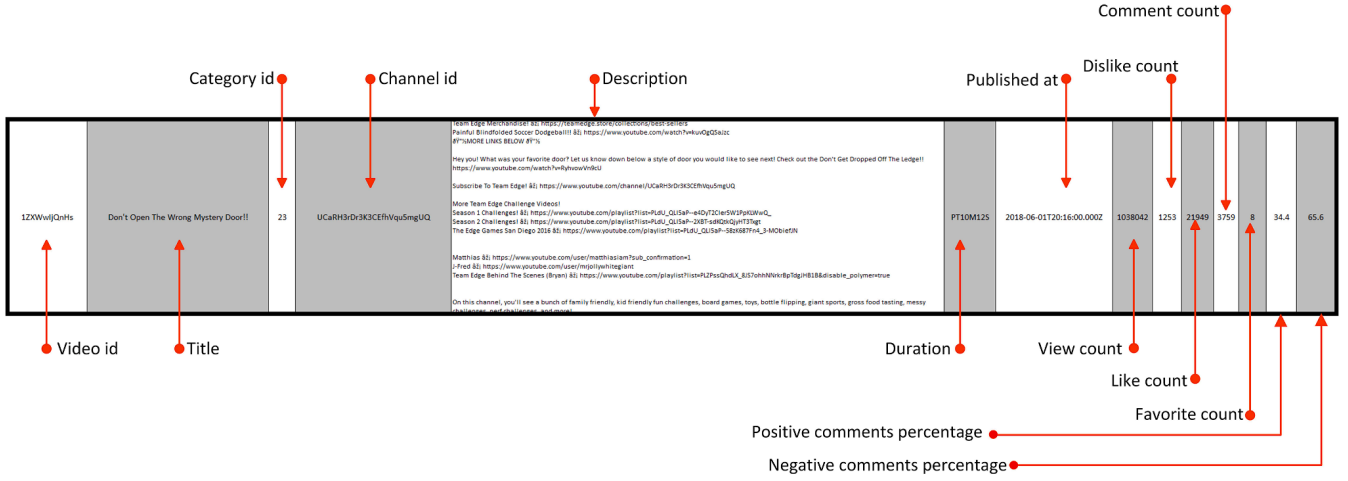


Figure 5: Sample dataset image. Make it in Powerpoint with svg images and save as pdf.
Sanity check: Zoom in and pixels should not break.

3.2. Detailed Methodology

Three paragraphs for explaining the pipeline and workflow of your study.
A figure depicting the pipeline only. If you use any equations, use the following as example as shown in Equation 1, which refers to theory of relativity [13] and Energy represented by E .

$$E = mc^2 \quad (1)$$

where E is the energy, m is the mass, and c is the speed of light.

3.3. Evaluation Metrics

One paragraph or more explaining the evaluation metrics with equations for each metric, where possible. The metrics should be defined specific to your dataset and problem and not generic (Accuracy w.r.t. your problem and NOT as ratio of true positives, etc.).

3.4. Experimental settings

One paragraph for experimental settings of your and competing methods (if any). (Optional) One paragraph for hyper-parameter settings and network architecture

A table with hyper-parameter settings (shown in Table 2) and a figure for network architecture (shown in Fig 6).

Table 2: Configuration table showing the network configuration of FCN used in this study. The table shows the various configuration settings used for FCN8.

Network Configuration	
Epochs	50
Learning rate	0.0001
Mini batch size	20
Optimizer	SGD
Momentum	0.9
Weight decay	0.0002
L_2 Regularization	None
Samples in training set	8498
Samples in validation set	786

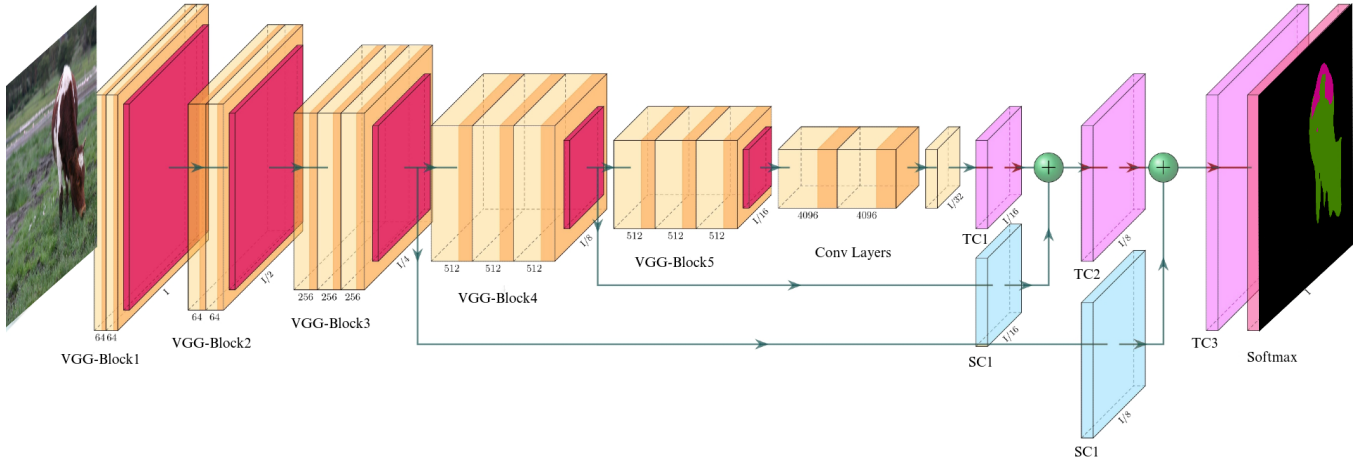


Figure 6: Sample network architecture image. Make it in Powerpoint with svg images and save as pdf. Sanity check: Zoom in and pixels should not break.

157 4. Results

158 One paragraph for summarizing your overall results.

159 4.1. Semantic Evidence Fusion

160 One/two paragraphs for all four figures/tables.

161 4.2. Fusion Strategy Optimality

162 One/two paragraphs for all four figures/tables.

163 *4.3. Uncertainty & Conflict Modelling*

164 One/two paragraphs for all four figures/tables.

165 *4.4. Redundancy vs Complementarity*

166 One/two paragraphs for all four figures/tables.

167 *4.5. Prior-Aware Semantic Fusion*

168 One/two paragraphs for all four figures/tables.

169 *4.6. Robustness Under Semantic Perturbations*

170 One/two paragraphs for all four figures/tables.

171 This section must contain only results and nothing else (not your own
172 opinion or any sort of discussion on quality of results). Sample shown in
173 Figure 7 and in Table 3).

Table 3: Performance comparison table showing the performance of various quantization techniques applied on various networks. The results are directly taken from the papers after applying quantization on the mentioned datasets.

Paper Name	Dataset Used	Performance Comparison							
		PA	MA	IoU	Time	Test Error	Sparsity Induced	size reduction	memory reduction
Vanhoucke et al. [26]	×				10x speedup				
Courbariaux et al. [27]	MNIST, SVHN, CIFAR-10					0.59%, 14.82%, 4.95%			
Gupta et al. [28]	MNIST, CIFAR-10					0.90%			
Denton et al. [29]	ImageNet 2012					0.83%	13%		
Lin et al. [30]	MNIST, SVHN, CIFAR-10					1.29%, 12.08%, 2.48%			
Hwang et al. [31]	MNIST					1.08%			
Anwar et al. [32]	MNIST					0.92%	17.1%		
Shin et al. [33]	TIMIT Corpus, MNIST					2.11%			
Xu et al. [39]	MICCAI Gland 2015	90.1%						6.4x	
Hubara et al. [46]	MNIST, SVHN, CIFAR-10, ImageNet 2012, Penn-tree bank					1.40% 2.53% 10.15%			
Lin et al. [47]	CIFAR-10					6.74%			20%
Proposed Approach	Pascal VOC 2012	89.30%	75%	56%		1.72%	40%	6.35x	84%

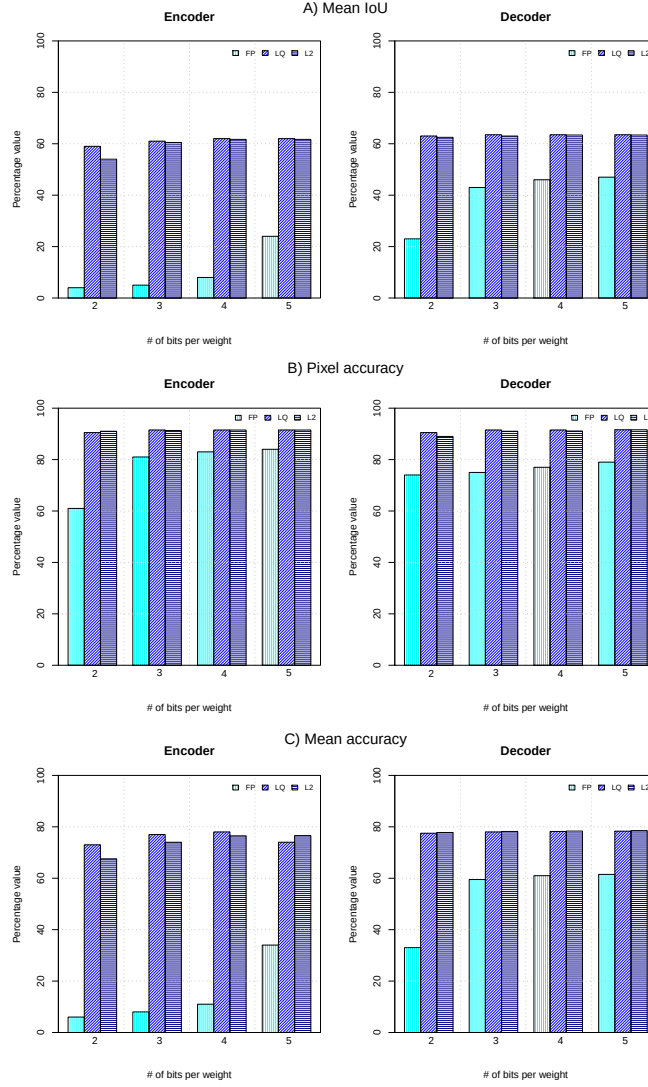


Figure 7: Sample Figure comparing the three quantization techniques Fixed Point (FP), Lloyd's quantizer (LQ) and L_2 error minimization (L_2) on the three performance metrics divided into encoder and decoder layers. Mean IoU is shown for the three techniques in Panel A), pixel accuracy in Panel B), and mean accuracy in Panel C) respectively. Note that FP is consistently worse than both LQ and L_2 , while L_2 and LQ are of comparable accuracy. Also, FP is most sensitive to number of bits in all metrics while L_2 and LQ are relatively insensitive.

174 5. Discussion

175 At least one paragraph for each research question. Your opinion on how
176 good/bad the results are. Draw inferences from the results here. Explain
177 novelty of your contributions and what was missing that you have explored
178 here. Discuss how results from your proposed method compare with other ex-
179 isting contemporary methods in this section. Any other point you would like
180 to discuss related to this study. Any assumptions that effect your analysis?

181 5.1. Limitations

182 One paragraph discussing the limitations of your work, if any?

183 5.2. Future Directions

184 One paragraph for what are the future directions in your opinion for
185 continuing this study.

186 6. Conclusion

187 One paragraph related to conclusions drawn from your whole experimen-
188 tation.

189 Declarations

190 The authors used the generative AI tool ChatGPT (OpenAI, San Fran-
191 cisco, CA, USA) to improve the language and clarity of the manuscript. The
192 authors reviewed and edited all content generated by the tool and take full
193 responsibility for the final version of the manuscript.

194 References

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